

MINI PROJECT

Multi-Factor Face-Based Application Authentication System

Using OpenCV, Python, LBPH Face Recognition

Project Documentation

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Submitted To

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1.Aim

To develop FaceGuard, a multi-factor face-based application authentication system that provides secure access to protected Windows applications using application-specific PIN verification and registered face recognition with liveness detection, while maintaining local security logs.

2.Algorithm

Compressed Algorithm

1. **Start FaceGuard.**
2. **Register user** with username, app-specific PIN, and face samples.
3. **Train and save** the user's LBPH face model.
4. **Select protected application.**
5. **Check user permission** for the application.
6. **Verify application PIN.**
7. **Perform liveness detection** using blink and head movements.
8. **Match face** with the registered face model.
9. If all checks pass, **grant access and launch the application.**
10. Otherwise, **deny access and record the attempt in security logs.**
11. **End.**

3.code

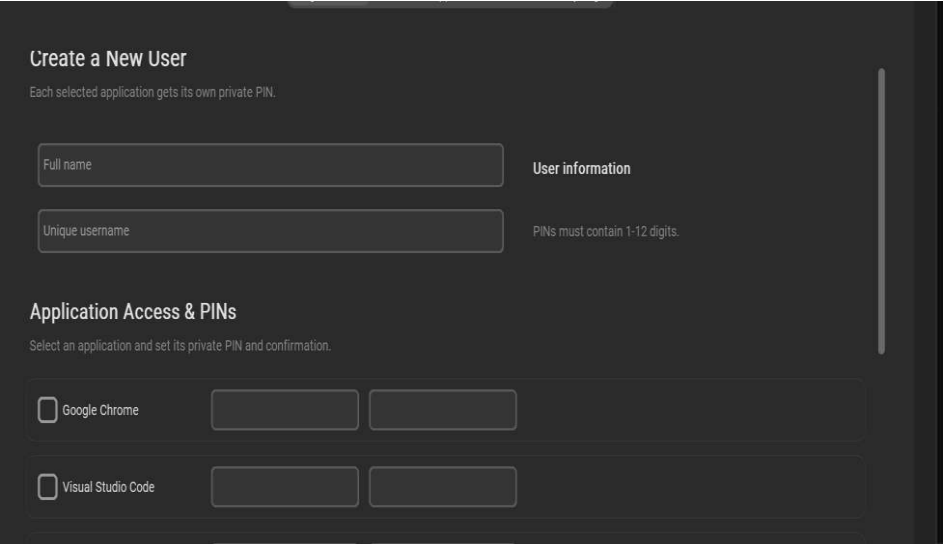
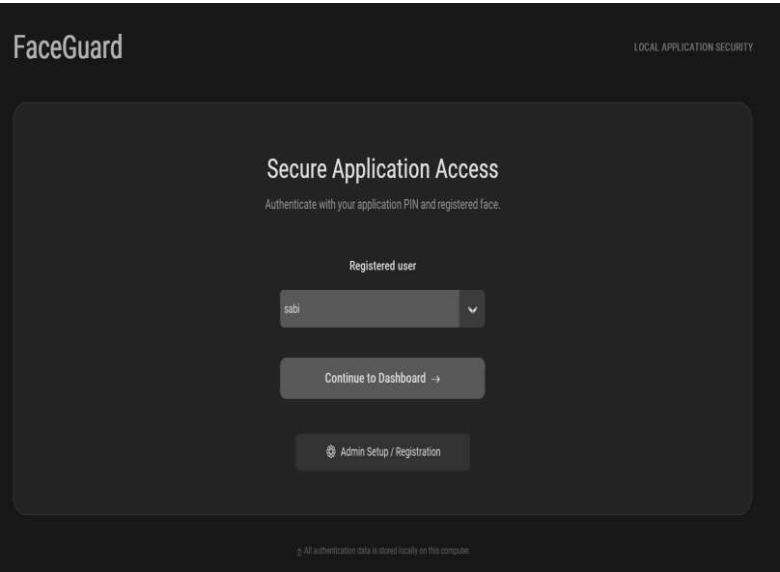
</> Python

```
# Verify application PIN
pin_hash = get_user_app_pin(user_id, app_id)
if not pin_hash or not verify_pin(entered_pin, pin_hash):
    add_security_log(username, app_name, "FAILED", "NOT_CHECKED", "DENIED")
    return

# Verify registered face with liveness detection
model = get_face_model(user_id)
face_result = verify_user_face(model["model_path"])

if face_result:
    add_security_log(username, app_name, "SUCCESS", "MATCH", "GRANTED")
    launch_application(app_path)
else:
    add_security_log(username, app_name, "SUCCESS", "NO_MATCH", "DENIED")
```

Output





5.Goal Of Project

The main goal of **FaceGuard** is to provide **secure access to protected Windows applications** by combining:

- **Application-specific PIN authentication**
- **Face recognition**
- **Liveness detection**
- **User application permissions**
- **Security logging**

This ensures that only the **authorized registered user** can access the selected application, while unauthorized attempts are detected and recorded.

5.Novelty

The novelty of **FaceGuard – Multi-Factor Face-Based Application Authentication System** lies in combining several security mechanisms into a single **local Windows application-access system**.

1. **Multi-Factor Application Authentication** – Access requires both an **application-specific PIN** and **registered face verification**.

2. **Application-Specific PINs** – Each protected application can have a **different PIN**, instead of using one common password.
3. **Liveness Detection** – The system checks **blink and head movements** before accepting face authentication, helping reduce simple photo-based spoofing.
4. **User-Application Binding** – Face authentication is performed against the **selected registered user's face model**, while database permissions determine which applications that user can access.
5. **Local Privacy-Focused Architecture** – Face models, PIN hashes, user information, and logs are maintained **locally using SQLite**, without depending on cloud authentication.
6. **Security Logging** – Every authentication attempt records the **PIN result, face result, application, date, time, and final decision**.
7. **Secure PIN Recovery** – A forgotten application PIN can be reset only after **registered-face verification and application-permission validation**, rather than displaying or recovering the old PIN.